Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack engineer with 2+ years shipping **production web applications**, **real-time workflows**, and **scalable backend services** using React, TypeScript, Python, SQL, Docker, and cloud-native CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **full-stack data interfaces** and backend services for 500+ users, optimizing SQL/indexing strategies to reduce dashboard latency from **1.2s to 350ms** under collaborative usage.
- Implemented **FastAPI orchestration** for 5K+ monthly workflows with validation, retry handling, and release safeguards, enabling reliable feature iteration across cross-functional product and engineering teams.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **React + Flask product features** handling 10K+ daily requests, translating UX requirements into reusable components, secure APIs, and maintainable backend modules in Agile sprints.
- Improved release velocity and quality via **CI/CD automation**, reducing turnaround time by **60%** while preserving stability for high-traffic grading and feedback user workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **Python experimentation pipelines** for 100K+ logs, producing structured diagnostics and reproducible evaluations to accelerate iterative product decisions on AI-driven interactive behaviors.
- Collaborated with researchers and engineers to convert analysis findings into actionable implementation plans, improving cross-team communication and reducing ambiguity in technical design tradeoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **API and ETL microservices** with Python/Java, AWS Glue, and S3, improving response time by **30%** and strengthening reliability for distributed data workflows.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito, boosting CI pipeline efficiency by **68%** and reducing technical debt across production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built **MERN modules** with REST APIs and WebSockets supporting eight workflows and 2M+ synchronized records, improving scalability and responsiveness of customer-facing platform operations.
- Automated **Docker/Jenkins deployment** workflows, reducing deployment time by **50%** and improving release consistency across iterative product launches and maintenance cycles.

Skills

Languages: **Python**, **JavaScript/TypeScript**, **Java**, SQL, C++
Frontend: **React.js**, HTML5, CSS3, reusable component systems, responsive UI
Backend: **Flask**, **FastAPI**, Node.js, REST APIs, microservices, PostgreSQL
Cloud/Infra: AWS (S3, Glue, RDS), Azure SQL, Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD
Engineering: Agile/Scrum, code reviews, testing automation, observability-first debugging

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a **real-time collaboration-ready fintech platform** supporting 50K+ users and 1M+ transactions with secure integrations, resilient backend services, and scalable deployment pipelines.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Designed a **schema-aware SQL assistant** across 10+ databases with validated query generation, structured summaries, and cross-platform tooling for faster developer and analyst workflows.

PiXMod | *Next.js, Python, MongoDB, Stripe, Transformers*

- Developed an **AI-powered full-stack SaaS** for image workflows with semantic retrieval, segmentation, and billing integration, processing 1K+ assets through production-ready APIs and storage layers.